

Joris Koefler

RESEARCH INTEREST

Positive Geometry and its applications in QFTs

EMPLOYMENT

- 2026 -2027** **Institute for Advanced Study**, Princeton, US
- Research Fellowship
 - School of Natural Sciences
 - Mentor: Nima Arkani-Hamed
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EDUCATION

- 2023 - 2026** **MPI for Mathematics in the Sciences**, Leipzig, GER
- PhD in Mathematics (3-year program)
 - Nonlinear algebra group
 - Advisor: Rainer Sinn
 - Thesis: Positive Geometries in Physics
- 2022 - 2023** **University of Warwick**, Coventry, UK
- M.Sc. in Mathematics (1-year program) - First Class
 - Advisor: Diane Maclagan
 - Thesis: A computational approach for tropical solutions to enumerative problems
- 2020 - 2022** **Ruhr University Bochum**, Bochum, GER
- B.Sc in Mathematics (3-year program) - Excellency Award
 - Advisor: Christian Stump
 - Thesis: Combinatorial models for the positive Grassmannian with a view towards Amplituhedra
- 2018 - 2020** **Ruhr University Bochum**, Bochum, GER
- B.Sc. in Physics (3-year program) - First Class
 - Advisor: Evgeny Epelbaum
 - Thesis: Application of a subtractive renormalisation scheme in Nucleon-Nucleon scattering in Effective Field Theory
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TEACHING

- 2024 - 2024** **Leipzig University**, Leipzig, GER
- Teaching Assistant
 - Combinatorial Commutative Algebra
- 2019 - 2022** **Ruhr University Bochum**, Bochum, GER
- Teaching Assistant
 - Mathematical Methods for Physicists I + II
 - Classical Theoretical Physics
- 2019 - 2021** **Ruhr University Bochum**, Bochum, GER
- Lab Supervisor
 - Biophysics
 - Geophysics

Research

- 2026 -** **Arrangement of Hyperplanes and a Quadric**
ongoing
- 2026 -** **Hodges Matrix, MHV Gravity Amplitudes, and Pentad Varieties**
With Emanuele Ventura, Hongmiao Yu
ongoing
- 2026** **Positive Geometry of Yang-Mills Correlators**
With Mattia Arundine, Veronica Calvo Cortes, Facundo Rost
- 2026** **Positive Genus Pairs from Amplituhedra**
With Dmitrii Pavlov and Rainer Sinn
[arXiv](#)
- 2025** **Connection Matrices in Macaulay2**
With Paul Görlach, Anna-Laura Sattelberger, Mahrud Sayrafi, Hendrik Schroeder, Nicolas Weiss, Francesca Zaffalon
To appear in JSAG, [arXiv](#)
- 2025** **Taking the Amplituhedron to the limit**
With Rainer Sinn
[arXiv](#)
- 2024** **Uniqueness of MHV Gravity Amplitudes**
With Umut Oktem, S. Paranjape, J. Trnka and B. Zecovic
Le Matematiche - Special Volume on Positive Geometry
[arXiv](#)

AWARDS

- 2023** **International Max-Planck Research School scholarship**
Max-Planck Institute for Mathematics in the Sciences

- 2022** **Excellency Award for Bachelors in Mathematics**
Ruhr University Bochum
- 2017** **UDE Scholarship for Nano-Engineering**
University Duisburg-Essen
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ORGANIZED EVENTS

- 2025** **Chow Lectures 2025 by Nima-Arkani Hamed**
MPI MIS Leipzig, [Link](#)
- 2024** **IMPRS COMBO III**
MPI MIS Leipzig, [Link](#)
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TALKS & PRESENTATIONS

- 2026** **Oberseminar Algebra**
Osnabrück, Germany
Adjoint and Positive Geometries of Positive Genus
- 2026** **Amplitudes Seminar**
Bonn, Germany
Positive Geometries of Positive Genus
- 2025** **SIAM: Conference on Applied Algebraic Geometry**
Madison Wisconsin, USA
Taking the amplituhedron to the limit
- 2025** **The Amplituhedron: Structure, Combinatorics, and Positive Geometry**
Les Diablerets, Switzerland
Taking the amplituhedron to the limit
- 2025** **Summer School in Total Positivity and Quantum Field Theory**
Harvard CMSA, USA
Taking the amplituhedron to the limit
- 2025** **The 11th Conference of the Fachgruppe Computeralgebra**
MPI MIS, Germany
Connection matrices in Macaulay2
- 2025** **Geometry Seminar**
University of Oslo, Norway
Taking the amplituhedron to the limit

- 2024** **Special Seminar on Positive Geometry**
MPI MIS, Germany
Uniqueness of MHV gravity scattering amplitudes
- 2024** **Meeting of Scientific Advisory Board**
MPI MIS, Germany
Positive Geometries, [Slides](#)
- 2022** **Seminar on Combinatorics**
Ruhr University Bochum
Combinatorial Models of the Grassmannian, [Link](#)